Structural and dynamical analysis of an integrated human/virus metabolic model provides insight to new treatment strategies against Covid-19

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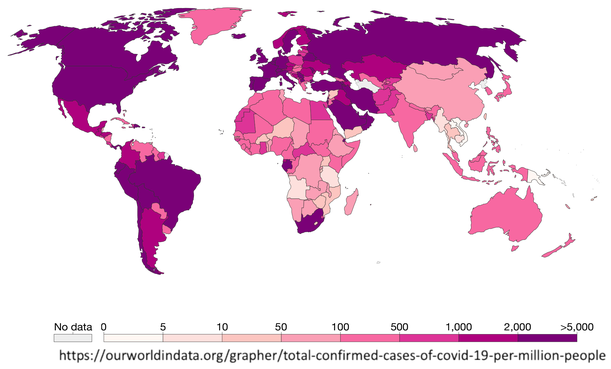
INTRODUCTION

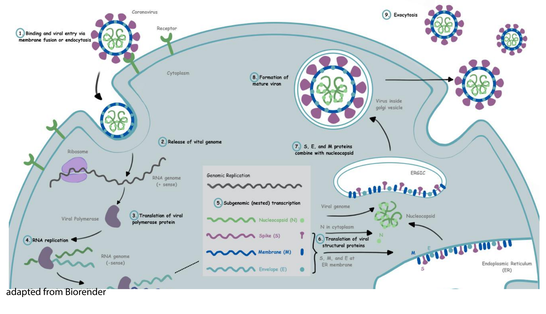
SARS-COV-2, the causative agent of the new coronavirus disease 2019 (COVID-19), belongs to a group of viruses commonly known as β-coronaviruses. These viruses are associated with mild to fatal cases of respiratory infections around the world. Previous outbreaks were caused by the MERs and SARS-2002 β-coronaviruses. However, the current outbreak, caused by SARS-COV-2, is the most severe due to difficult containment strategies and the lack of treatment regimens.



The COVID-19 disease, which originated in Wuhan, China is currently responsible for over 400 thousand deaths in 216 countries across the world (https://www.who.int/emergencies/diseases/novel-coronavirus-2019).

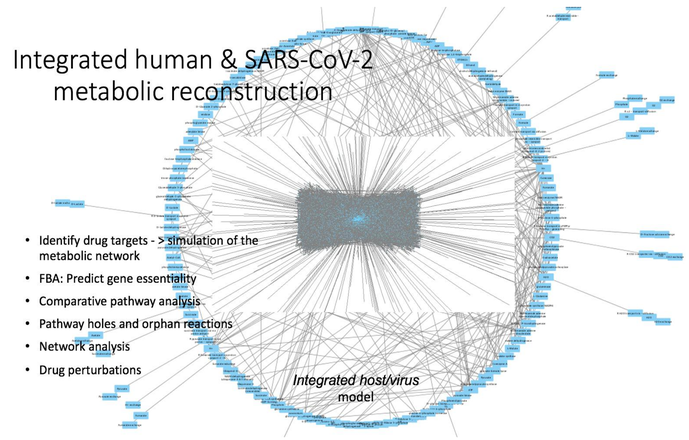
BACKGROUND

The mechanism of replication of the SARS-CoV-2 virus into the human cell involves an initial binding and attachment of the virus' spiked glycoprotein to the ACE2 receptor of its host. During endocytosis, the genetic material of the virus is injected into the human cell, where it loses its protective envelope. The virus, now ready for replication, is released into the nucleus of the cell. The replication, assembly and maturation of the viral proteins leads to cell death and a proliferation of the virus within the human body.



The absence of FDA approved drugs against the new SARS-CoV-2 virus has prompted the urgent need to design new drugs and treatment management strategies against COVID-19.

METHODS



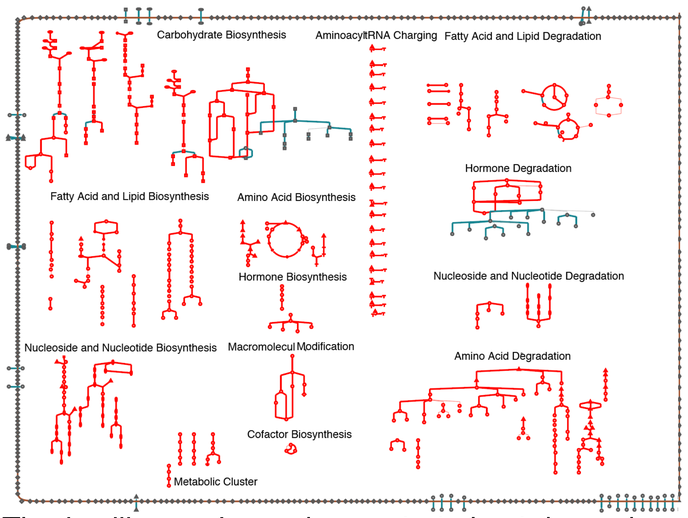
First, we developed a genome scale metabolic model that integrates the human airway epithelial cell with the SARS-CoV-2 virus using the methods outlined in Aller et al. 2018. We have compared our model with a previously constructed integrated model of the human macrophage and the SARS-CoV-2 virus (Renz et al. 2020) .

Secondly, we use Flux Balance Analysis and Flux variability Analysis to assess the models and to make predictions.

Third, we used a new tool *findCPcli* (Bannerman et al. 2020; Oarga et al. 2020) to predict potential new drug targets.

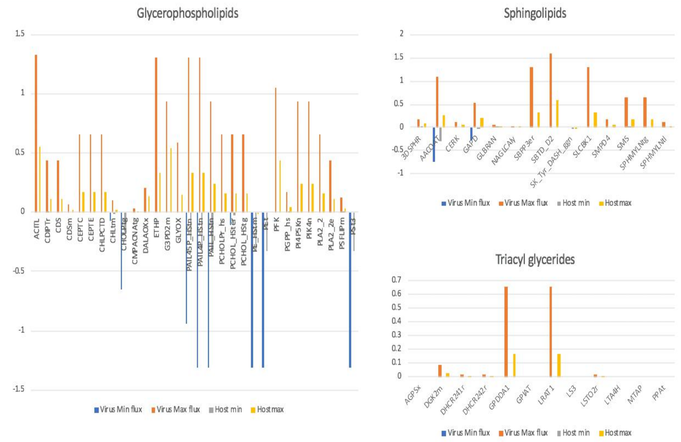
RESULTS

Our results demonstrate the biochemical requirements for the growth and general maintenance of the human cell in their healthy states. The entry of the SARS-CoV-2 virus into the cell causes biochemical and structural changes in the host cell leading to a change of cell functions or cell death.

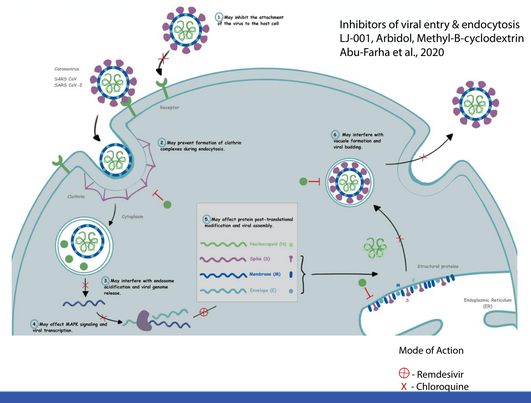


The *in silico* analyses demonstrate host dependency factors within various metabolic pathways that are required for SARS-COV-2 infection, replication, survival and viability.

RESULTS (Contd)



We have highlightedf 813 metabolic reactions, in the human body that have been hijacked as essential for the virus' proliferation and viability.



CONCLUSION

We have prioritised 15 reactions in lipid metabolism, coenzyme transport and metabolism, energy production and conversion, amino acid and nucleotide transport and metabolism as potential drug targets.

Our results provide insight into treatment management strategies to inhibit viral entry into the cell and insights to immune regulation and antioxidant therapy.

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